

anywhere, and they came to be concentrated in towns in the north and Midlands of England that rapidly grew into major industrial centers as the century progressed.

Social changes

Huge numbers of workers were drawn to these new cities, which became synonymous with poor living and working conditions for the workforce, many of whom were children. This influx led to the creation of an urban underclass. It took a long time before the workers saw any improvement in their lives, and the realization that they should share in the rewards of this social and economic transformation, rather than simply be exploited as mere drudges, came very slowly. In the meantime, however, the increasingly wealthy factory owners emerged as a significant political voice.

The wider world

As late as 1860, Britain was, by some margin, still the world's leading industrial and mercantile power, but other Western nations were quick to see how they could benefit. In continental Europe, industrialization was initially uncertain, inhibited by the kind of political instability Britain had managed to avoid, such as the revolutions of 1848. Later, the pace of its development would rival Britain's. In 1840, Germany and France each had around 300 miles (480 km) of railroad lines; in 1870, both had 10,000 miles (16,000 km). Similarly, pig-iron output

from each rose from about 125,000 tons in 1840 to 1 million in 1870.

However, the most startling developments came in the United States, where there were around 3,300 miles (5,300 km) of railroad in 1840, almost all in the northeast. By 1860, this had increased to 32,000 miles (51,500 km), and by 1900 it had soared to 193,000 miles (310,600 km) of rail track. The production of pig iron rose similarly: in 1810, it was a little less than 100,000 tons a year; in 1850, it was approaching 700,000 tons; in 1900, it was over 13 million.

The role of steel

By about 1870, in both Europe and the United States, a second wave of industrialization began, in which oil, chemicals, electricity, and steel became increasingly important. The production of steel had been transformed after 1855, when English engineer Henry Bessemer devised a way to make the metal lighter, stronger, and more versatile; from that point forward, steel would prove the linchpin for industry. In 1870, total world steel production was 540,000 tons, but within 25 years it had risen to 14 million tons, and railroads, armament production, and the shipbuilding industry all benefited from its ready availability.

While Germany was beginning to threaten Britain's preeminent industrial position in Europe, quadrupling its industrial output between 1870 and 1914, the United

Isambard Kingdom Brunel

No figure better encapsulates the determination, ambition, and vision that drove the first phase of the Industrial Revolution in Britain than the prodigiously hard-working Isambard Kingdom Brunel (1806–1859). He was responsible for an extraordinary series of firsts: the world's

longest bridge (the Clifton Suspension Bridge), the world's longest tunnel (Box Tunnel in Wiltshire), and the world's largest ship (the *Great Eastern*). In 1827, still only 21, he was appointed chief engineer of the Thames Tunnel. In 1833, he became engineer to the newly formed Great Western Railway, which by 1841 linked London directly with Bristol, whose docks he had rebuilt from 1832. Believing it should be

possible to travel directly from London to New York, Brunel also designed the world's first practical ocean-going steamship, the *Great Western*. He followed this with the screw-driven iron-built *Great Britain*. Despite his great vision, delays and cost overruns dogged many of Brunel's projects, but his works include some of the grandest feats of engineering the world had yet seen.